

Question			Marking details	Marks available																									
				AO1	AO2	AO3	Total	Maths	Prac																				
6	(a)		Indicative response: Before jumping, all of the energy is in the form of gravitational potential energy (GPE). As the jumper falls, the original GPE is transferred to KE. When the cord starts to stretch the KE continues to increase initially, as does the elastic potential energy (EPE). Both GPE and KE will decrease as the cord stretches further, until eventually the KE becomes (momentarily) zero at the lowest point of the jump. At this point the decrease in GPE is equal to the EPE in the cord. As the jumper starts to move upwards the EPE is transferred back to KE and GPE. The jumper does not reach the initial height above the ground, as some energy is transferred to air molecules (and / or reference to hysteresis). Additional comments: Some energy is transferred to air molecules. Explanation of why jumper does not reach original height after first bounce. N.B. Table below is to aid marking – candidates must write in extended prose to gain 6 marks.	2	4		6																						
			<table><tr><th>Position</th><th>E_g</th><th>E_k</th><th>E_{el}</th></tr><tr><td>At start (before jump)</td><td>Max or mgh where h is height above ground</td><td>0</td><td>0</td></tr><tr><td>Free fall, cord slack</td><td>Decreasing</td><td>Increasing</td><td>0</td></tr><tr><td>Cord stretching</td><td>Decreasing</td><td>Continues to increase initially before decreasing.</td><td>Increasing</td></tr><tr><td>At bottom of first bounce</td><td>Minimum - don't accept zero unless explained. Accept $-mg\Delta x$ where Δx is distance fallen</td><td>0</td><td>Maximum or $\Delta E_{el} = -mg\Delta x$</td></tr></table>	Position	E_g	E_k	E_{el}	At start (before jump)	Max or mgh where h is height above ground	0	0	Free fall, cord slack	Decreasing	Increasing	0	Cord stretching	Decreasing	Continues to increase initially before decreasing.	Increasing	At bottom of first bounce	Minimum - don't accept zero unless explained. Accept $-mg\Delta x$ where Δx is distance fallen	0	Maximum or $\Delta E_{el} = -mg\Delta x$						
Position	E_g	E_k	E_{el}																										
At start (before jump)	Max or mgh where h is height above ground	0	0																										
Free fall, cord slack	Decreasing	Increasing	0																										
Cord stretching	Decreasing	Continues to increase initially before decreasing.	Increasing																										
At bottom of first bounce	Minimum - don't accept zero unless explained. Accept $-mg\Delta x$ where Δx is distance fallen	0	Maximum or $\Delta E_{el} = -mg\Delta x$																										